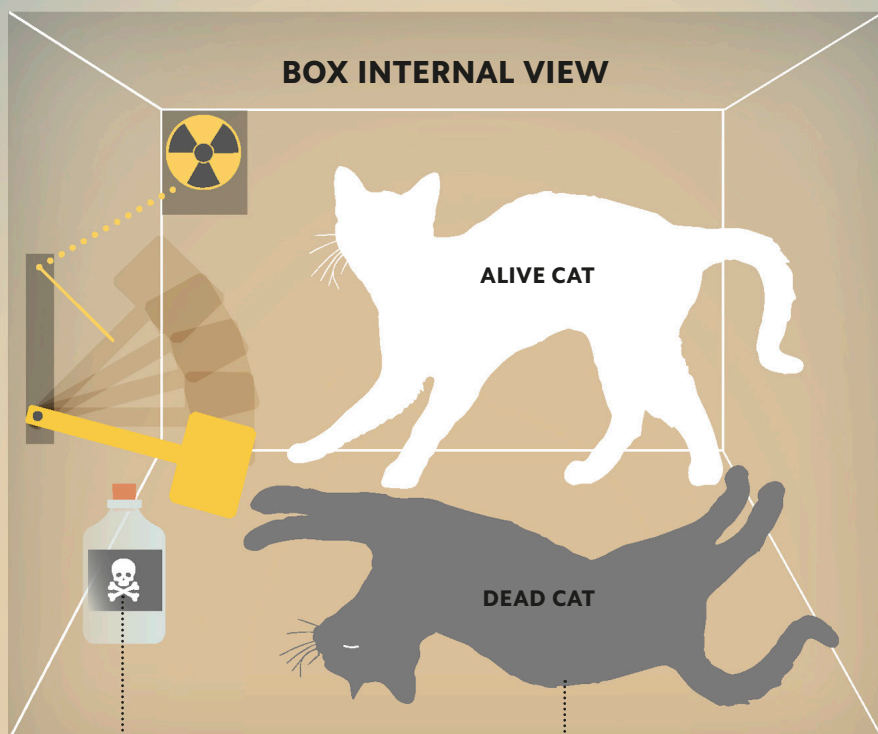




**RADIOACTIVE
DECAY**

The random and unpredictable nature of radioactive decay means that, until an observation is made, it is uncertain whether or not the poison has been released.

**CAT
CONUNDRUM**

Without looking inside the box, it is impossible to know if the cat is alive or dead, so it can be thought to be in a superposition of being alive and dead at the same time.

PARADOX IN A BOX

The mystery of what occurs before wave function collapse inspired a famous thought experiment known as “Schrödinger’s cat.” In this thought experiment, a cat in a box could be killed at any time by poison released via nuclear decay. In one interpretation of quantum mechanics (see pp.52–53), the superposed states of dead and alive exist until an observer opens the box. Erwin Schrödinger devised this thought experiment to highlight the absurdity of a cat being both dead and alive until someone checks on it.